



The Algorithmic Age: America's Choice for AI Leadership

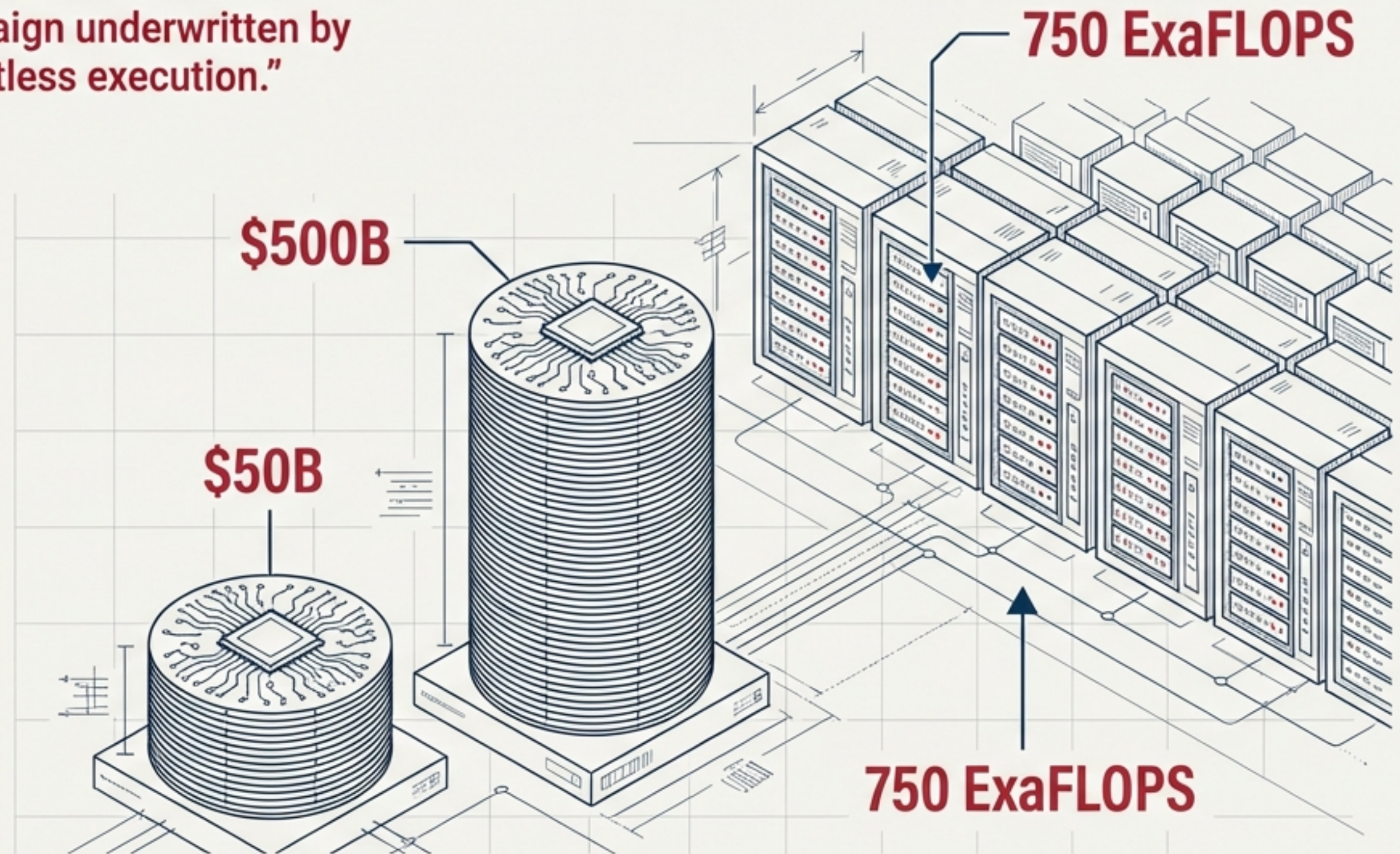
The United States is the global leader in Artificial Intelligence. To ensure the economic and national security benefits accrue in the United States and the technology is developed and deployed responsibly, we must take active steps to safeguard our current leadership position.

The Global Arena: A New Era of Strategic Competition

“Beijing is unmistakably forging the infrastructure of an AI-driven future... a state-orchestrated campaign underwritten by colossal capital and driven by relentless execution.”

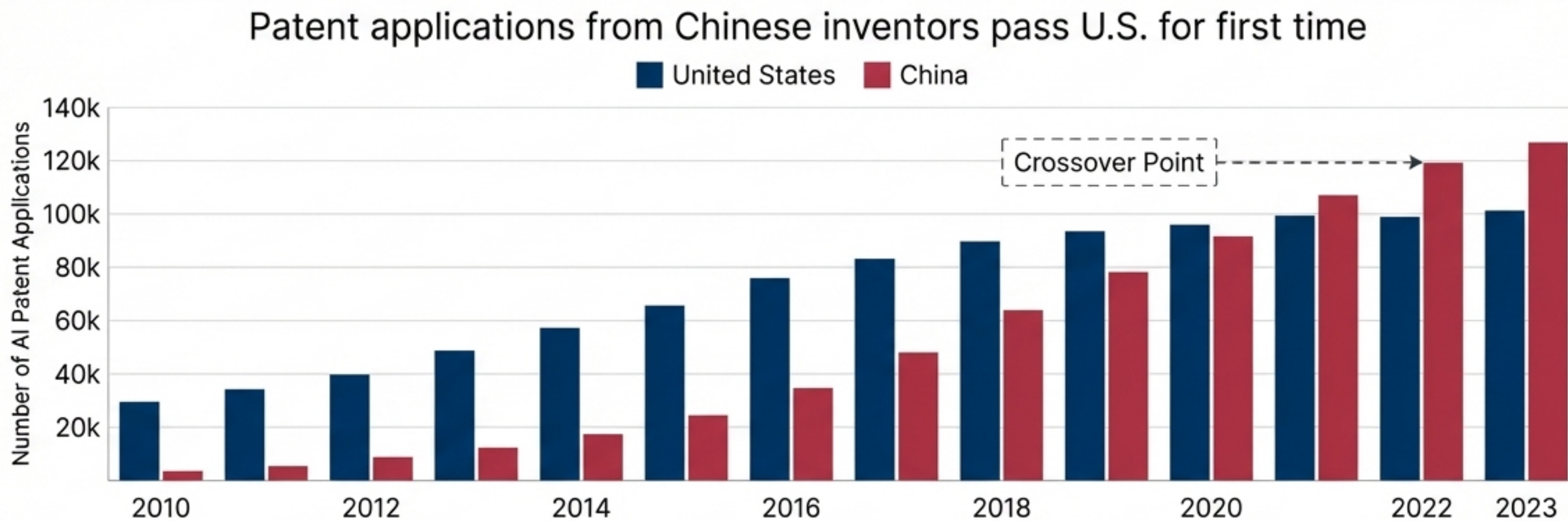
China's 'AI+' Initiative

- **Goal:** Undisputed global leadership by 2030.
- **Investment:** Foundational semiconductor fund nearing **\$50 billion**.
- **R&D Budget:** **Half a trillion** dollars annually.
- **Infrastructure Target:** **750 ExaFLOPS** of compute power.



Source: Bajraktari Testimony

While America Pioneers, China Is Rapidly Closing the Gap



Source: Axios chart, data from House AI Task Force Report

- The Chinese Communist Party is aggressively narrowing this lead by matching U.S. AI performance.
- China is outpacing the U.S. in the number of AI patent applications, journal publications, and journal citations.
- China is mass-producing STEM talent and systematically targeting American intellectual property via sophisticated cyber intrusions.

The American Paradox: We Invent the Future, but Risk Adversaries Industrializing It

“Our Achilles’ heel is not invention, but adoption: the translation of breakthrough research into scaled industrial and government capabilities.”



INVENTION

**THE
ADOPTION
GAP**



SCALED ADOPTION



Brittle supply chains for critical hardware.



Bureaucratic inertia stifles agility and procurement.



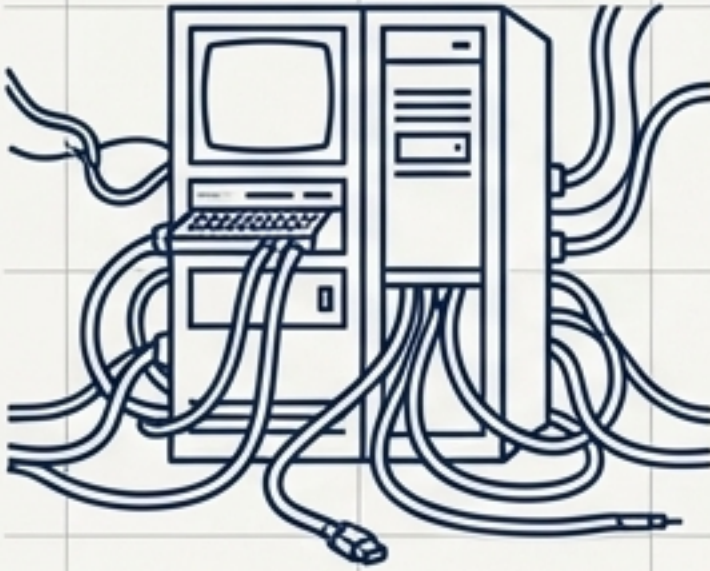
A 'patchwork' of fragmented policies undermines long-term investment.



Organizational culture of risk aversion stifles experimentation.

Inside the Machine: Overcoming Barriers to Federal AI Adoption

The Department of Defense and federal agencies face widespread opportunities for AI, but systemic hurdles hinder enterprise-wide deployment.



Legacy Systems

Many agencies' antiquated IT infrastructure is simply unfit for the demands of modern AI...

80% of the government's \$100B+ IT spend goes to operating existing legacy systems.



Talent Gaps

A chasm exists in data science, machine learning, and AI ethics expertise...

Roles and associated AI knowledge and skills are unclear and highly varied across the federal workforce.



Procurement & ATO

Federal acquisition processes are often too slow and risk-averse to procure cutting-edge AI capabilities... The ATO process needs to be more efficient.

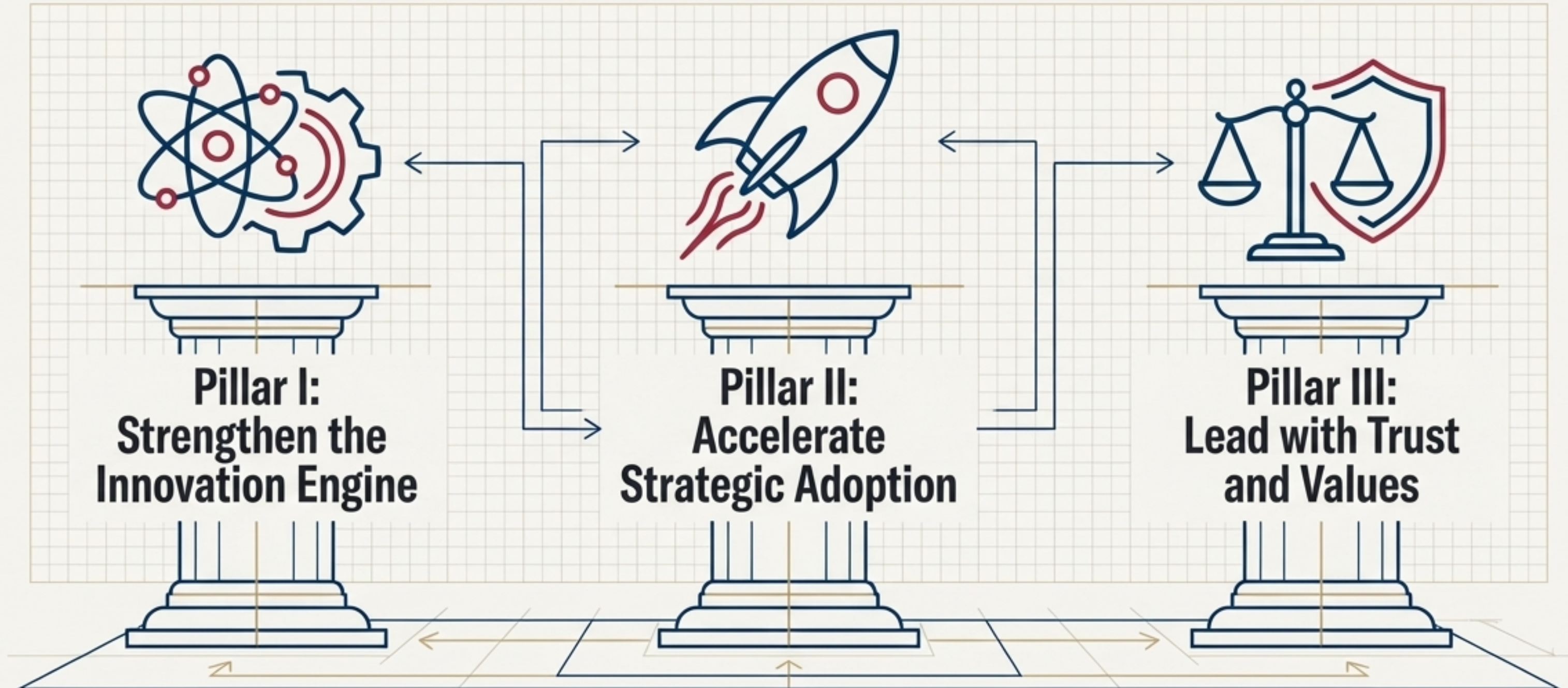


Data Silos

Federal data, the lifeblood of AI, is too often of poor quality, siloed, and poorly governed.

A National Blueprint for American AI Leadership

This report articulates guiding principles, 66 key findings, and 89 recommendations... It is intended to serve as a blueprint for future actions that Congress can take to address advances in AI technologies.



Underlying Principle: Keep Humans at the Center of AI Policy.



Pillar I: Strengthen the Innovation Engine

“Federal investments in fundamental research have enabled the current AI opportunity... The private sector performs the majority of U.S. R&D, but the federal government is the leading source of support for basic research.”

Federal R&D Expenditures for AI (FY2024 Requested)

NSF: \$757.6M	NIH: \$925.7M
DOE: \$240.6M	DARPA: \$322.1M

Strategic Actions

- ➡ Double non-defense AI R&D to **\$32 billion** annually.
- 🌐 Fully operationalize the **National AI Research Resource (NAIRR)** to connect researchers with computational data, software, and training models.
- 🌉 Increase technology transfer from university R&D to market to bridge the “valley of death”.
- 🤝 Promote public-private partnerships to avoid redundant research and accelerate commercialization.



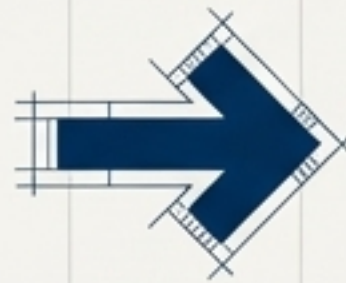
Pillar II: Accelerate Strategic Adoption in Government

AI is a critical component of national security. For the U.S. government, the integration of AI at scale is not a discretionary upgrade—it is a strategic imperative.



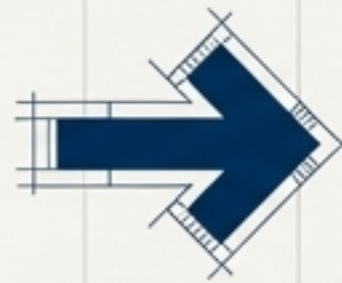
Enable Experimentation

Support experimental environments and “proof of concept” pilot projects to understand capabilities before enterprise deployment.



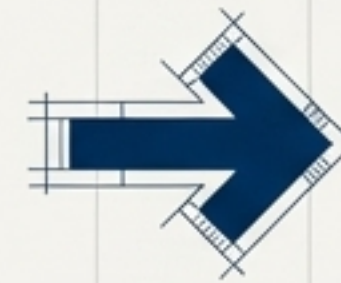
Scale Secure Platforms

Accelerate the scaling of DoD-secured platforms like **NIPRGPT** and **CamoGPT** to provide secure alternatives to commercial models.



Streamline Deployment

Make the **Authority to Operate (ATO)** process more efficient and harmonize policies across the DoD.



Empower Leadership

Empower Chief AI Officers to drive adoption across their agencies and utilize AI to reduce administrative burdens.

Pillar II: Accelerate Strategic Adoption Across the Economy

The U.S. has a significant gap in the appropriate talent needed to research, develop, and deploy AI applications—and this gap is growing.



Strategic Actions - The Talent Offensive

- Invest in **K-12 STEM and AI Education** and broaden participation, as U.S. is lagging.
- Support **new pathways** into the AI workforce beyond traditional degrees, such as an “AI Scholarship for Service” program.
- Standardize work roles, job categories, and skill sets for AI-related jobs.



Strategic Actions - Empowering Small Business

- Provide resources for small business AI literacy and adoption.
- Ensure small businesses have access to capital and AI resources (data, compute) to compete with larger firms.
- Ease regulatory compliance burdens that disproportionately affect small businesses.



Pillar III: Lead with Trust and American Values

The federal government should foster trust by adopting responsible principles and policies that capture the benefits of AI while addressing its risks and leading by example.

Protect Civil Rights & Liberties



- Require **'humans in the loop'** for highly consequential decision-making.
- Empower sectoral regulators to address AI-related risks in housing, employment, and finance.

Safeguard Data Privacy



- Ensure privacy laws are generally applicable and **technology-neutral**.
- Explore mechanisms like privacy-enhancing technologies to promote access to data for research.

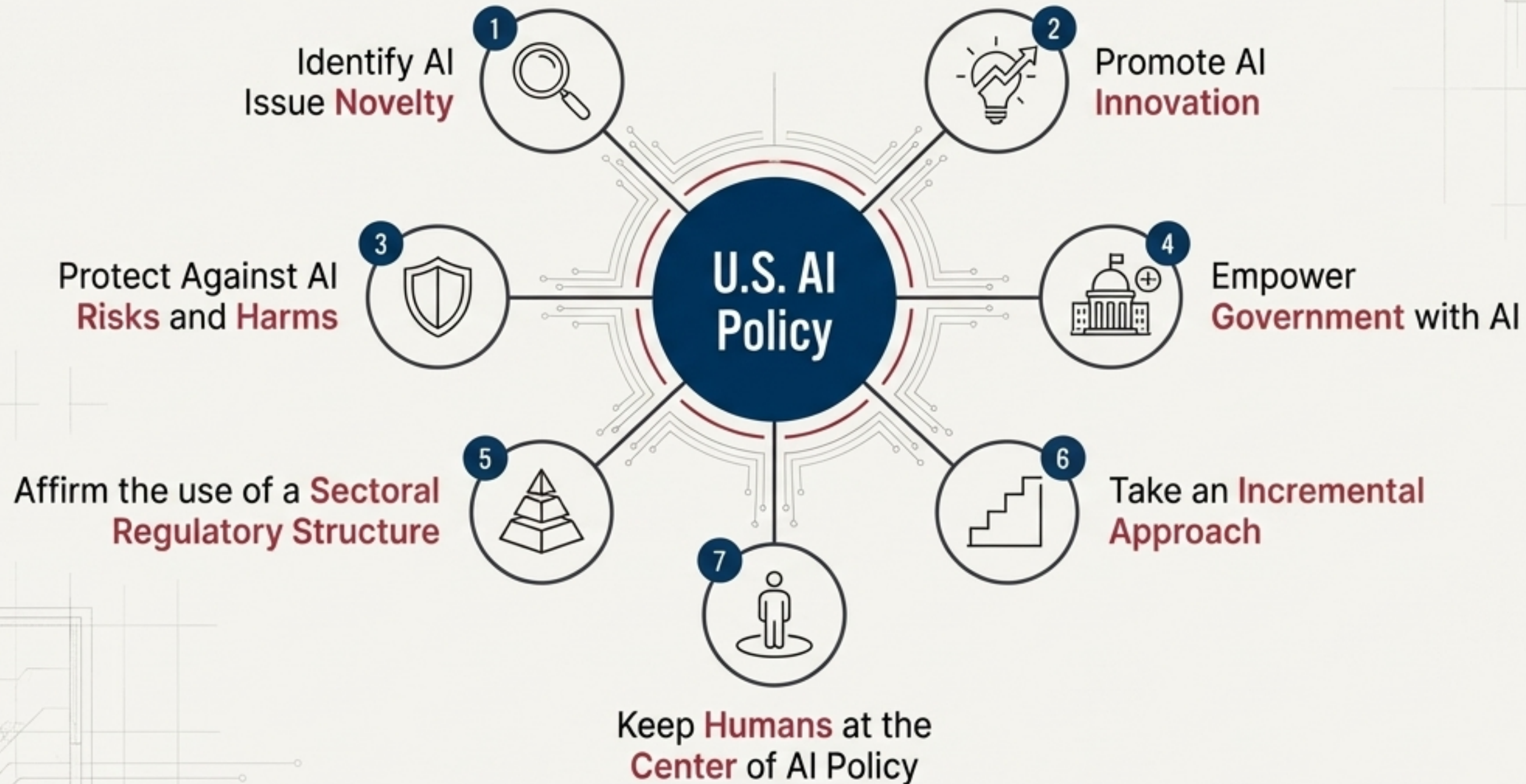
Ensure Content Authenticity



- Support a risk-based, multipronged approach to content authenticity.
- Address demonstrable harms of synthetic content, not speculative ones.

The Guiding Philosophy for a National AI Strategy

The **Bipartisan House AI Task Force** adopted several high-level principles to frame its policy analysis. These represent considerations that transcend specific AI issues and can help guide future congressional efforts.



Source: Bipartisan House Task Force on Artificial Intelligence Report

From Blueprint to Reality: An Agile, Sector-Specific Approach

Healthcare



Challenge: "Lack of ubiquitous, uniform standards for medical data impedes system interoperability."

Action: Support standards to facilitate data sharing while protecting patient privacy.

Financial Services



Challenge: "Small financial services firms can be at a disadvantage in AI adoption."

Action: Ensure regulations do not impede small firms from adopting AI tools.

Energy



Challenge: "The growing demands of AI are creating challenges for the grid."

Action: Promote broader adoption of AI to enhance energy infrastructure, production, and efficiency.

Agriculture



Challenge: "Lack of reliable network connectivity in rural communities impedes AI adoption."

Action: Assess programs to advance AI in precision agriculture and address infrastructure gaps.

An Agenda for Action: Securing American AI Supremacy

Pillar I: Strengthen the Innovation Engine

- Commit to sustained, large-scale federal investment, aiming to double non-defense AI R&D to **\$32 billion** annually.
- Fully resource and operationalize the **National AI Research Resource (NAIRR)**.
- Aggressively protect American IP and counter critical tech-focused espionage.

Pillar II: Accelerate Strategic Adoption

- Execute a comprehensive **National Talent Offensive**, from K-12 education to reforming STEM immigration.
- Create a **National Reserve Digital Corps** and a **U.S. Digital Service Academy**.
- Streamline regulations for critical infrastructure and turbocharge the **"Rip and Replace"** program for untrusted network equipment.

Pillar III: Lead with Trust and Alliances

- Establish a high-level **Technology Competitiveness Council (TCC)** at the White House to orchestrate a unified strategy.
- Forge techno-democratic alliances for secure networks and construct new frameworks like a **"Five Eyes" for sensitive tech**.
- Champion an **'AGI for Peace'** initiative to shape global norms.

The Choice in the Algorithmic Age



“The United States faces a generational AI challenge, where supremacy will dictate our prosperity, security, and influence this century. Today’s decisions determine whether America masters this algorithmic domain by our values or if others define it for us. By embracing AI, rebuilding our foundations, and reigniting innovation, America can and must secure its technological preeminence and enduring security.”

— Ylli Bajraktari, Testimony before the House Foreign Affairs Committee

This is a complex, yet vital choice, echoing President Kennedy’s call, to lead.